JSX

Interview Q&A; — Topic 2

8 Questions • Beginner → Expert

● Beginner ● Intermediate ● Expert

Interview Questions & Model Answers

1 Beginner Q1. What is JSX?

➜ JSX is a syntax extension for JavaScript that looks like HTML but is compiled by Babel into React.createElement() calls. The browser never sees JSX — it sees plain JavaScript objects that form the Virtual DOM.

■ *Interview Tip: Always mention: JSX → Babel → React.createElement() → JS Object → Virtual DOM*

2 Beginner Q2. Is JSX mandatory in React?

➜ No. You can write React.createElement() directly. But JSX makes it far more readable. In React 17+, you don't even need to import React at the top of every file — the new JSX transform handles it automatically.

■ *Interview Tip: Bonus: Mention React 17+ automatic JSX transform — shows you know the evolution.*

3 Beginner Q3. Why className instead of class in JSX?

➜ Because JSX is JavaScript under the hood, and class is a reserved keyword in JS for defining classes. To avoid conflicts, React uses className for CSS classes. Similarly, for is replaced with htmlFor in labels.

■ *Interview Tip: Always pair className with htmlFor — interviewers often ask both together.*

4 Intermediate Q4. Why can't you use if statements inside JSX?

➜ JSX curly braces {} only accept JavaScript expressions — things that evaluate to a value. if is a statement, not an expression. So we use ternary operators (condition ? a : b) or short-circuit && instead.

■ *Interview Tip: Expression = produces a value. Statement = performs an action. Know this distinction cold.*

5 Expert Q5. What is $$typeof and why does it exist?

➜ It's a Symbol that React attaches to every element for XSS security. Since JSON cannot serialize Symbols, a malicious server-injected object can never fake being a valid React element — React will refuse to render it.

■ *Interview Tip: This is a senior-level question. Most candidates don't know it — saying this will impress.*

6 Intermediate Q6. What does Babel do with JSX?

➜ Babel is a JavaScript compiler that transforms JSX syntax into React.createElement() function calls that browsers can understand. Without Babel (or a similar tool like SWC), browsers would throw a syntax error on JSX.

■ *Interview Tip: SWC is the newer, faster Rust-based alternative to Babel used in Vite/Next.js — good to mention.*

7 Beginner Q7. What is a Fragment and why use it?

➜ A Fragment (<> or) lets you return multiple elements from a component without adding an extra DOM node. This keeps the DOM clean and avoids unnecessary wrapper divs that break CSS layouts like Flexbox/Grid.

■ *Interview Tip: Mention the short syntax <> vs — Fragment supports key prop, <> does not.*

8 Beginner Q8. Can you use JavaScript expressions inside JSX? Give examples.

➜ Yes — anything inside {} is evaluated as a JS expression. Examples: {name} for variables, {isLoggedIn ? : } for conditionals, {items.map(i => {i})} for lists, {getGreeting()} for function calls.

■ *Interview Tip: The .map() for lists example is extremely common in interviews — always have it ready.*